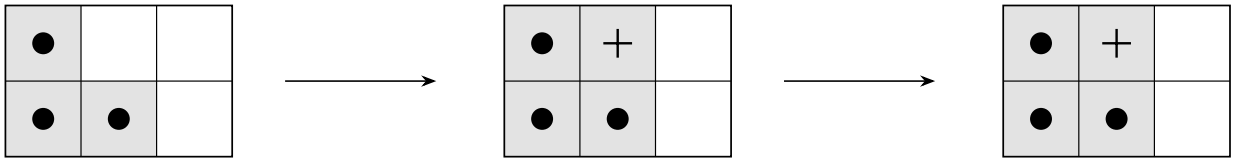
Week 98 / Growing from a few seeds / Grades 4–5

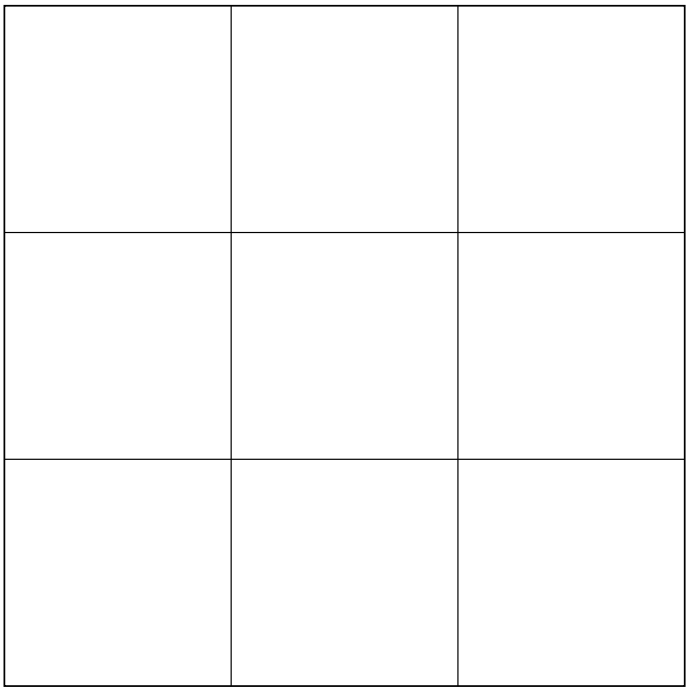
Seeds are the squares filled at the start. Put counters on them. Add a counter to an empty square when at least two squares sharing its sides are filled. Your partner points to two filled neighbors before each addition. Counters stay put. Stop when no more can be added. Diagonal touches do not count. Opposite edges do not meet; there are no squares outside the board.

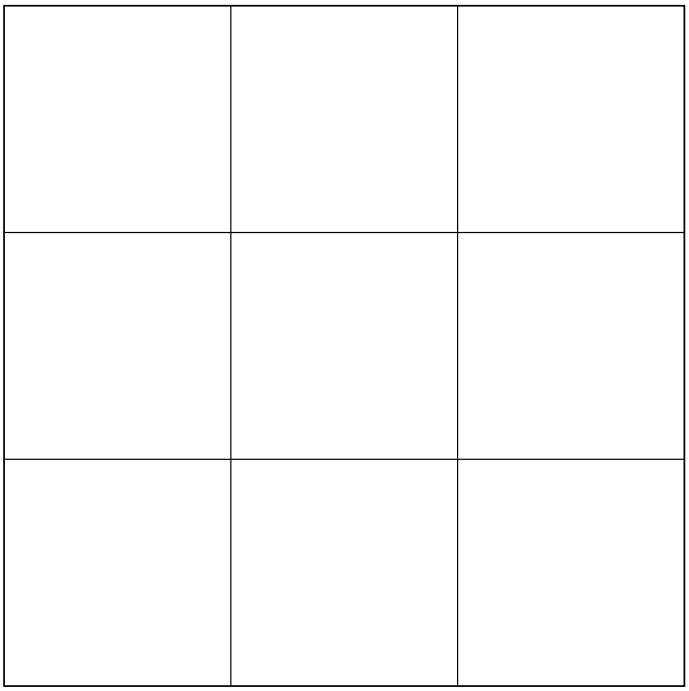
Mark seeds with • and added squares with $+$.



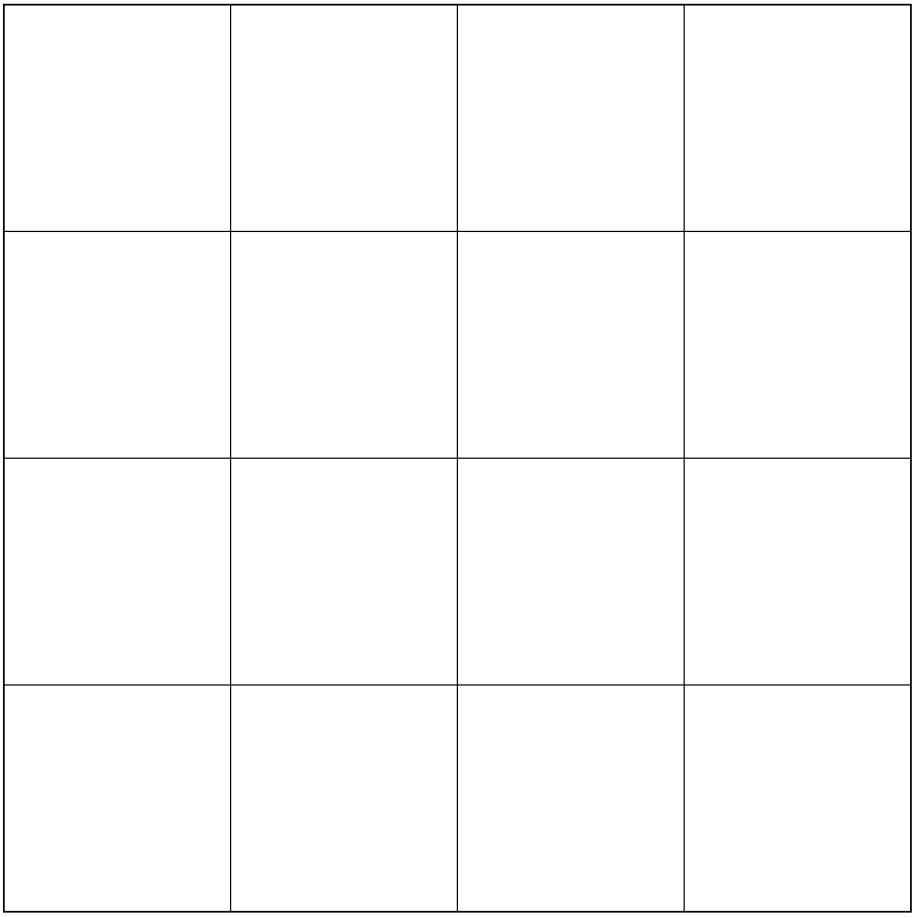
Start One square added Stopped

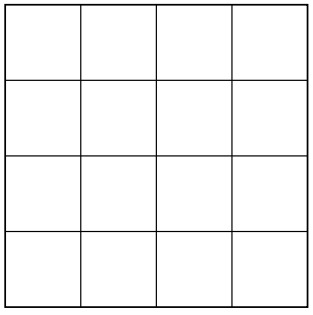
Problem 1: Choose exactly three seeds on a 3-by-3 board. How many different final numbers of filled squares can you make?

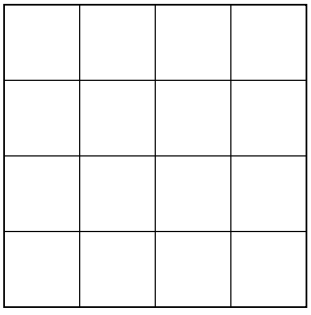


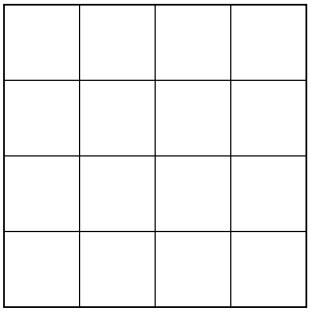


Problem 2: What is the smallest number of seeds you can use to fill a 4-by-4 board? Find three different starts using your best number.









Problem 3: Find the fewest seeds you can use to fill a 5-by-5 board and a 10-by-10 board. Grow these boards by drawing • for seeds and + for additions instead of using counters.

